University of CINCINNATI
CEAS

Attendin

CEAS '26 EXPO

The Problem: Outdated Methods

Many educational institutions and organizations still rely on manual sign-in sheets, verbal roll calls, or basic digital forms to track attendance. These traditional methods are highly inefficient, wasting valuable instruction and administrative time. More importantly, they are prone to human error and highly susceptible to academic dishonesty, such as students checking in for absent peers. This reliance on outdated tracking results in inaccurate records and a lack of reliable data for assessing actual student engagement, creating a clear need for a secure, automated solution.

The Process: Geolocation Verification

When an instructor creates an attendance session via the web dashboard, they establish a secure, localized geofence around the classroom. To check in, students open the Attendin mobile app to prompt the system to retrieve their device's current location. Our core algorithm then calculates the relative distance between the student's location and the center of the established geofence. If the student's device falls within the allowed radius during the active class window, their attendance is instantly verified and synced to the Cloud Firestore database. If they are outside the radius, the check-in is securely blocked.

- **Session Creation:** Faculty set the time, location, and geofence radius.
- **Distance Calculation:** The system compares the student's location against the geofence center.
- **Automated Verification:** Attendance is granted or denied in real-time based on physical proximity.

Figure 1: The Attendin mobile interface, displaying the secure login, student dashboard, and active session check-in screen.

Analysis: Cost & Feasibility

A core advantage of the Attendin architecture is its highly cost-effective and scalable design. For the capstone development, the project budget is exactly $0, utilizing the free tiers of our core technology stack: GitHub for version control, Flutter for cross-platform front-end development, and Google Firebase for authentication and database management.

When analyzing the feasibility of a production-level rollout scaled for a standard university, we estimate the operating cost to be approximately $180 per month. The bulk of this cost is allocated to Google Firebase for hosting and managing up to 1TB of active user data. This demonstrates that Attendin is not only technologically feasible but also financially viable as a scalable enterprise solution.

- **Development Cost:** $0 (Leveraging free tiers of Flutter, Firebase, and GitHub).
- **Production Estimate:** ~$180/month for a single-university scale.
- **Tech Stack:** Flutter (Front-End) & Google Firebase / Cloud Firestore (Back-End).
- **Scalability:** Architecture natively supports high volumes of concurrent check-ins.

Abstract

Traditional attendance tracking relies on manual sign-in sheets or verbal roll calls. These methods are time-consuming, prone to human error, and susceptible to academic dishonesty. Attendin is a streamlined, automated system that leverages geolocation technology to verify physical presence, ensuring that attendees are actually in the room before checking in.

The system is split into two simple tools: a cross-platform mobile app for students and a comprehensive web dashboard for instructors. Our ultimate goal with Attendin is to cut down on administrative busywork for faculty, ensure the integrity of attendance records, and provide a seamless, modern user experience that seamlessly integrates into the daily academic routine.

System Design & Interfaces

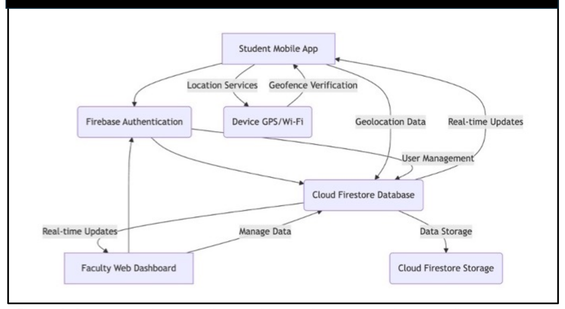

Figure 2: System architecture diagram illustrating the real-time data flow between the student mobile app, Google Firebase backend, and the instructor web dashboard.

Final Outcomes & Results

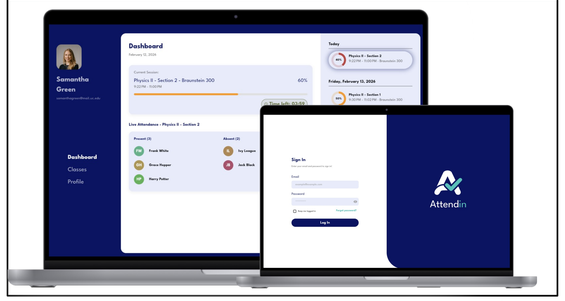

Figure 3: The comprehensive faculty web dashboard, allowing instructors to seamlessly manage attendance sessions and monitor real-time roster updates.

Built with a focus on cross-platform accessibility and ease of use, Attendin's architecture utilizes a dual-interface approach to seamlessly serve both students and faculty. The student-facing mobile application, developed using the Flutter framework, provides a highly responsive and intuitive experience tailored for both iOS and Android devices. Through this interface, students can effortlessly view their upcoming class schedules, monitor their historical attendance records, and securely check into active sessions using on-device GPS verification.

Conversely, instructors and administrators access the system through a robust web dashboard engineered for comprehensive classroom management. This portal allows faculty to dynamically generate localized geofences by defining specific coordinates and radius parameters to create active attendance sessions. The web interface also equips administrators with advanced management tools, including the ability to manually override attendance statuses and export comprehensive data reports.

Bridging these two distinct interfaces is a secure and highly scalable Google Firebase backend. Firebase Authentication ensures that all user credentials are rigorously verified before granting access. Furthermore, the infrastructure relies on Cloud Firestore to guarantee that every student check-in, geolocation calculation, and roster update is synchronized in real-time across the entire network, ensuring absolute data integrity.

The Attendin platform has been successfully completed, delivering a fully functional and rigorously tested attendance tracking system. The finalized student-facing mobile application, built with Flutter, provides a seamless and intuitive check-in experience across both iOS and Android devices. Simultaneously, the comprehensive faculty web dashboard is fully operational, allowing instructors to effortlessly create and manage secure geofenced sessions. Through extensive testing, our core geolocation algorithm and Google Firebase backend have proven to be accurate and reliable, ensuring real-time database synchronization without compromising user security. Ultimately, Attendin successfully achieves its primary goal: providing a modern, cost-effective, and fraud-resistant solution that entirely eliminates the administrative burden of traditional attendance tracking.

- **Cross-Platform Success:** Mobile app fully deployed and functional on iOS and Android.
- **Real-Time Sync:** Firebase backend successfully handles concurrent data updates without latency.
- **Accurate Verification:** Leverages device-native location APIs to retrieve high-accuracy GPS coordinates, calculating the precise distance vector against the active session's geofence radius.
- **Cost Optimization:** Implemented local data caching and optimized Firebase queries to drastically reduce cloud usage, keeping operational costs near zero.
- **Goal Achieved:** Delivered a secure, cost-optimized, and user-friendly alternative to manual roll calls.

Broader Impacts

- **Reducing Wasted Class Time:** By automating the data collection process, the system reclaims valuable instructional time that is currently lost to manual roll calls or passing around sign-in sheets.
- **Restoring Academic Integrity:** Current systems like static QR codes are easily exploited by students, sign in sheets can sometimes be marked off by friends, by enforcing physical presence through geolocation verification, this ensures that attendance records accurately reflect a student's actual participation.
- **Modernizing Classroom Infrastructure**: This project replaces outdated, insecure methods with a secure, scalable digital architecture. By utilizing the devices students already own, this project demonstrates that secure, verifiable attendance tracking can be implemented at minimal additional cost to the university.

Challenges & Future Scope

- **Schema-First Architecture:** We established a database schema early on, which allowed us to work on the UI skeletons separately. Once the frontend was finished, we were able to implement Firebase smoothly without having to rework our code.
- **High-Volume Stress Testing:** To guarantee the system could handle an entire lecture hall checking in simultaneously, we wrote automated seeding scripts. These injected massive volumes of mock interactions into Firebase, allowing us to stress-test the dashboard's real-time listeners and ensure UI responsiveness without physically gathering large test groups.
- **FERPA Compliance & Privacy:** To meet the strict legal constraints of a university environment, we engineered a privacy-first database schema. The system deliberately stores only check-in statuses and timestamps, ensuring no continuous or historical location data is ever recorded.

Future iterations will focus on developing an API bridge to seamlessly integrate with Learning Management Systems like Canvas for automatic gradebook synchronization. Additionally, we plan to incorporate biometric authentication like FaceID to further harden the system against fraud.

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